

**Quiz (Habanero)****Q1.** What is the standard form of a quadratic function?

- (A)  $ax^2 + bx + c$   
 (B)  $ax + bx^2 + c$   
 (C)  $ax^3 + bx^2 + c$   
 (D)  $ax^2 - bx + c$   
 (E)  $ax - bx^2 + c$

**Q2.** Which of the following is a quadratic function in standard form?

- (A)  $x^2 + 3x - 2$   
 (B)  $x^3 - 2x^2 + x - 1$   
 (C)  $x^2 - 4x + 4$   
 (D)  $x^4 + 2x^2 - 3$   
 (E)  $x - 2$

**Q3.** The quadratic function  $f(x) = 2x^2 + 5x - 3$  represents the!

- (A) temperature changes in a city  
 (B) rainfall in a region  
 (C) height of a projectile  
 (D) growth of a population  
 (E) depth of a canyon

**Q4.** What is the value of  $a$  in the quadratic function  $f(x) = ax^2 + 2x - 1$ , if the function represents the!

- (A) 0.5  
 (B) 1.2  
 (C) 2.5  
 (D) 3.1  
 (E) 4.2

**Q5.** Which of the following is NOT a key feature of a quadratic function?

- (A) vertex  
 (B) axis of symmetry  
 (C) y-intercept  
 (D) x-intercept  
 (E) period

**Q6.** The quadratic function  $f(x) = x^2 + 2x + 1$  has a!

- (A) minimum value of 0  
 (B) maximum value of 0  
 (C) minimum value of 1  
 (D) maximum value of 1  
 (E) minimum value of -1

**Q7.** What is the equation of the axis of symmetry of the quadratic function  $f(x) = x^2 + 4x + 4$ ?

- (A)  $x = -2$   
 (B)  $x = -1$   
 (C)  $x = 0$   
 (D)  $x = 1$   
 (E)  $x = 2$

**Q8.** The quadratic function  $f(x) = 3x^2 - 2x - 1$  represents the!

- (A) growth of a population  
 (B) height of a projectile  
 (C) depth of a canyon  
 (D) temperature changes in a city  
 (E) rainfall in a region

**Open Ended Questions (FRQ)****Q9.** Describe the key features of the quadratic function  $f(x) = 2x^2 + 5x - 3$ , and explain how it can be used to model the temperature changes in a city.**Q10.** Find the equation of the axis of symmetry of the quadratic function  $f(x) = x^2 + 4x + 4$ , and explain its significance in the context of the problem.**Chef's Tips**

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- Tip 1: Make sure students understand the definition of a quadratic function in standard form.
- Tip 2: Emphasize the importance of identifying key features of quadratic functions.
- Tip 3: Encourage students to use real-world examples, such as weather and environment, to illustrate the applications of quadratic functions.